

Project Title 2:

AI BANK LOAN REVIEW ASSISTANT

AI-Powered Bank Loan Review Assistant Using NLP and Generative AI

2. Business Problem

Customers seeking loans often compare multiple banks before making a decision. However, information regarding customer service quality, loan approval speed, processing efficiency, and customer satisfaction is scattered across thousands of online reviews and feedback platforms.

Customers face challenges such as:

- Difficulty comparing multiple banks.
- Lack of centralized customer feedback analysis.
- Time-consuming review reading process.
- Inability to quickly identify strengths and weaknesses of loan providers.

This results in inefficient decision-making and reduced customer confidence.

3. Proposed Business Solution

The proposed system is an AI-Powered Bank Loan Review Assistant designed to simplify the loan selection process for customers.

Today, customers often visit multiple bank websites, compare loan products, and read numerous customer reviews before deciding where to apply for a loan. This process can be confusing because each bank has different eligibility criteria, documentation requirements, approval timelines, interest rates, and customer service experiences.

The proposed solution uses Natural Language Processing (NLP) and Generative AI to analyze customer reviews, loan-related feedback, and bank-specific information. Instead of manually searching through large amounts of information, users can interact with a chatbot that provides quick, easy-to-understand insights.

The chatbot acts as a loan comparison and decision-support assistant by helping users:

- **Compare loan services across multiple banks.**
- **Understand customer experiences and satisfaction levels.**

- Identify common complaints and challenges faced by borrowers.
- Learn about typical documentation requirements.
- Understand loan approval timelines and service quality.
- Receive guidance on whether their available documents may meet common loan requirements.

The goal is not to approve or reject loans, but to help customers make more informed decisions before approaching a bank.

How the Solution Works

Bank Loan Reviews + Loan Policy Information



Data Cleaning & Processing



NLP Analysis

(Sentiment Analysis + Keyword Extraction)



Insight Generation



Generative AI Chatbot



Customer Questions & Answers

Example 1: Comparing Banks

Customer Query:

"Which bank has better customer service for home loans?"

Chatbot Response:

"Based on the analyzed customer reviews, Bank A receives more positive feedback regarding staff support, communication, and issue resolution. Bank B receives a higher number of complaints related to delayed responses during the loan process."

Example 2: Loan Approval Experience

Customer Query:

"Which bank provides faster home loan approval?"

Chatbot Response:

"Customers frequently mention that Bank C has a quicker approval process, with many reviews highlighting faster document verification and sanction procedures compared to other banks in the dataset."

Example 3: Common Customer Complaints

Customer Query:

"What are the most common complaints regarding home loans?"

Chatbot Response:

"The most frequently reported issues include delays in document verification, lengthy approval timelines, repeated requests for documents, and lack of communication during processing."

Example 4: Documentation Guidance

Customer Query:

"I have Aadhaar Card, PAN Card, salary slips, and bank statements. Are these documents generally sufficient for a home loan?"

Chatbot Response:

"Based on common loan requirements observed across the selected banks, Aadhaar Card, PAN Card, salary slips, and recent bank statements are frequently required documents. However, some banks may additionally request income tax

returns, employment verification, property-related documents, or proof of residence depending on the loan type and eligibility criteria."

Example 5: Customer Satisfaction Comparison

Customer Query:

"Which bank has the highest customer satisfaction for personal loans?"

Chatbot Response:

"According to customer review analysis, Bank D has the highest positive sentiment score. Customers particularly appreciate transparent communication, smooth documentation processes, and quick disbursement of loans."

Business Benefits

- Reduces the time customers spend researching loan options.
- Provides a centralized platform for comparing multiple banks.
- Helps customers understand real borrower experiences.
- Simplifies loan-related decision-making.
- Improves transparency by summarizing customer feedback.
- Demonstrates the practical application of NLP and Generative AI in banking and financial services.

One-Line Summary for Viva

"This project acts as an intelligent loan comparison assistant that analyzes customer reviews and loan-related information from multiple banks, enabling customers to make informed borrowing decisions through a simple conversational chatbot interface."

5. Scope of the Project

The system will:

- Analyze loan-related customer reviews.
- Compare selected banks based on review sentiment.

- Identify service quality indicators.
 - Summarize customer experiences.
 - Provide AI-generated responses through a chatbot interface.
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7. Tools & Technologies

Programming Language

- Python

Development Environment

- Jupyter Notebook

Data Processing

- Pandas
- NumPy

NLP Libraries

- NLTK
- TextBlob

Generative AI

- Gemini API

Visualization

- Matplotlib
- Seaborn

Dataset Source

- Bank Customer Review Dataset (Kaggle/Custom Dataset)
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